

Brief overview on how to start exploring **functionality**

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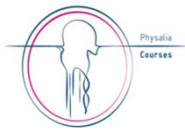
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OscarGenomics



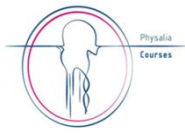
Previous steps to Functional Analysis



1. Perform your GWAS
2. Extract significant SNPs (after adjusting for all confounding covariates, pop strat and multiple test correction)
3. Obtain genes carrying the significant SNP or close by
4. Use a bioinformatic tool to obtain biological information on those genes



Functional analysis



- Examining the function of genes and their role in biological processes.
 - by looking at how changes in a gene's sequence affect its function
 - by studying how different genes interact with each other to carry out specific functions in the cell.
 - using bioinformatics tools to analyze large datasets of genomic information, such as the sequence of an entire genome or the expression levels of thousands of genes in different tissues or under different conditions. This can help identify patterns and connections between genes that may be involved in specific functions or processes, such as disease development or response to environmental stimuli.

Overall, a genomic functional analysis can provide valuable insights into the roles that genes play in biological processes, and can help researchers better understand the underlying causes of diseases and other biological phenomena



Work Flow in FUMA

- Download bkg genes: <https://www.ensembl.org/info/data/ftp/index.html>
- Results overview in FUMA
 - <https://fuma.ctglab.nl/snp2gene>
 - Input file: GWASresults.txt
- Variant Effect Prediction in Ensembl
 - <https://www.ensembl.org/Multi/Tools/VEP>
 - Input file (significant SNPs): Map.selected.rs
 - Output file (Functional Info): Select 'Gene' column
- Enrichment analysis
 - <https://fuma.ctglab.nl/gene2func>
 - Input files: GENES from significant SNPs ('Gene' column from VEP)
 - Background genes from the specie (Canis_familiaris.bkg_genes from [here](#))



Work Flow in DAVID



- Visit the DAVID website and create an account, if you don't already have one.

— <https://davidbioinformatics.nih.gov/>

- Log in to your account and go to the "Functional Annotation" section of the platform.
- Choose the appropriate algorithms and methods for your analysis. DAVID offers a variety of options for functional analysis, including clustering, enrichment analysis, and pathway mapping.

The screenshot shows the DAVID website's navigation menu. The 'Functional Annotation' section is highlighted with a red arrow. The menu items are as follows:

- Home
- Start Analysis
- Shortcut to DAVID Tools
- Technical Center
- Downloads & APIs
- Terms of Service
- About DAVID
- About LHRI

Documentation

- Getting Started
- Overview
- List Upload & Management
- Analysis Tools
 - Gene Search
 - Gene List Report
 - Gene Report
 - Functional Annotation Summary
 - Functional Annotation Table
 - Functional Annotation Chart
 - Functional Annotation Clustering
 - Filtered Gene List Report
 - Gene Functional Classification
 - Cluster Term Report
 - Related Genes
 - Related Terms
- Gene Search
 - Gene Search
 - Gene List Report
 - Gene Report
 - Functional Annotation Summary
 - Functional Annotation Table
 - Functional Annotation Chart
 - Functional Annotation Clustering
 - Filtered Gene List Report
 - Gene Functional Classification
 - Cluster Term Report
 - Related Genes
 - Related Terms
- Gene Functional Classification
- Ortholog
- Gene ID Conversion

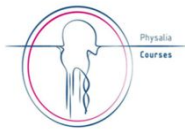
Current Version

DAVID 2025 (March 2026)

This update summarizes cumulative developments since the [2022 NAR publication](#).

- Introduced two new tools: **DAVID Ortholog** for cross-species functional analysis and **DAVID Gene Search** for identifier-agnostic gene exploration.
- Implemented a new backend architecture that decouples the frontend from the legacy Java processing engine using a Servlet layer and REST APIs, enabling asynchronous processing and improved performance.

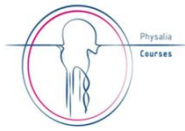
Work Flow in DAVID



- Visit the DAVID website and create an account, if you don't already have one.
 - <https://david.ncifcrf.gov/>
 - <https://davidbioinformatics.nih.gov/>
- Log in to your account and go to the "Functional Annotation" section of the platform.
- Choose the appropriate algorithms and methods for your analysis. DAVID offers a variety of options for functional analysis, including clustering, enrichment analysis, and pathway mapping.

The screenshot shows the DAVID Bioinformatics Resources website. The header includes the DAVID logo, the text "DAVID Bioinformatics Resources Laboratory of Human Retrovirology and Immunoinformatics (LHRI)", and the LHRI logo. The navigation bar contains links: Home, Start Analysis, Shortcut to DAVID Tools, Technical Center, Downloads & APIs, Term of Service, About DAVID, and About LHRI. The main content area is divided into two columns. The left column has a sidebar with "Overview" and "Functional Annotation" (which is highlighted by a red arrow). Below "Overview" is a list of tools: "The Database for Functional Annotation", "Gene Names Batch Viewer", "Gene Functional Classification", and "Gene ID Conversion". The right column has a "Hot Links" section and a "Multiple positions available in LHRI" section. The "Multiple positions available in LHRI" section contains three bullet points: (1) Scientist - Virology position available to perform the detective proviral study in our Basic Research Section, (2) Scientist - cytokines and HIV available in our Basic Research Section. We are looking for a cytokine immunologist who is interested in virus (HIV/HSV/KHSV) pathogenesis in myeloid immune cell types (macrophages, dendritic cells and microglia cells), and (3) Postdoctoral fellow available in our Basic Research Section. This position is an excellent opportunity for a young Ph.D. who has no experience in virus research and seeks a career in a new research field. You will learn how

Work Flow in DAVID

The screenshot shows the DAVID Gene Name Batch Viewer interface. It has a blue header with the DAVID logo, "Gene Name Batch Viewer", and "DAVID Bioinformatics Resources, NIAID/NIH". Below the header is a navigation bar with links: Home, Start Analysis, Shortcut to DAVID Tools, Technical Center, Downloads & APIs, Term of Service, About DAVID, and About LHRI. The main content area is titled "Gene Name Batch Viewer" and includes a "Submit your gene list to start!" button. There are also links to "Tell us how you like the tool", "Read technical notes of the tool", and "Contact us for questions". The interface is divided into four steps: Step 1: Enter Gene List, Step 2: Select Identifier, Step 2a: Select species, and Step 3: List Type. Step 4: Submit List is also visible. Red arrows point from the text on the right to the corresponding steps in the interface. The interface also includes a "Key Concepts of 'Search Related Genes'" section with a description of the tool's functionality and a "Find Related Genes Tool" section with a description of the tool's functionality.

Paste the list of genes (output from the getGenes.R)

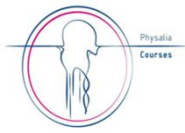
Select OFFICIAL_GENE_SYMBOL

Type the species

Submit analysis

Use the results of your functional analysis to gain insights into the biological processes and functions that are represented in your data. You can use the information from DAVID to help identify potential pathways and mechanisms involved in the development of diseases, for example, or to investigate how different genes or proteins interact to perform specific functions in the cell.

Work Flow in DAVID



DAVID Bioinformatics Database Functional Annotation Tool
DAVID Bioinformatics Resources, NIAID/NIH

Home Start Analysis Shortcut to DAVID Tools Technical Center Downloads & APIs Term of Service About DAVID About LHRI

Upload List Background

Gene List Manager

Select to limit annotations by one or more species [Help](#)

- Use All Species -
Canis lupus familiaris(7)

Select Species

List Manager [Help](#)

List_1

Select List to:
Use Rename
Remove Combine
Show Gene List

Annotation Summary Results [Help and Tool Manual](#)

Current Gene List: List_1
Current Background: Canis lupus familiaris
7 DAVID IDs
Check Defaults ☒ Clear All

☒ Functional_Annotations (6 selected)

☒ COG_ONTOLOGY	28.6%	2	Chart	
☒ UP_KW_BIOLOGICAL_PROCESS	42.9%	3	Chart	
☒ UP_KW_CELLULAR_COMPONENT	42.9%	3	Chart	
☒ UP_KW_MOLECULAR_FUNCTION	28.6%	2	Chart	
☒ UP_KW_PTM	28.6%	2	Chart	
☒ UP_SEQ_FEATURE	100.0%	7	Chart	

☒ General_Annotations (0 selected)

☒ Interactions (1 selected)

☒ Pathways (1 selected)

☒ KEGG_PATHWAY	42.9%	3	Chart	
☐ WIKIPATHWAYS	28.6%	2	Chart	

☒ Protein_Domains (0 selected)

Red annotation categories denote DAVID defined defaults

Combined View for Selected Annotation

Functional Annotation Clustering

Functional Annotation Chart

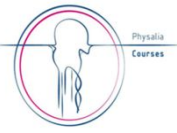
Functional Annotation Table

Go again to the Shortcut to **David Tools** and select '**Functional Annotation**'

You can select the different options to gain insights into the biological processes and functions that are represented in your data. You can use the information from DAVID to help identify potential pathways and mechanisms involved in the expression of the phenotype. For example, to investigate how different genes or proteins interact to perform specific functions in the cell.



Limitations



- FA relies on algorithms and methods that make assumptions and simplifications about the data being analyzed. These assumptions may not always hold true in all cases, and they can affect the accuracy and reliability of the results.
- FA are often based on large datasets that may not be representative of all possible scenarios. This can lead to bias in the results, and can make it difficult to generalize the findings to other situations or organisms.
- FA can provide valuable insights, but it is important to carefully consider the limitations and uncertainties of the methods and algorithms used, and to interpret the results with caution.

